Result 8.7

Three distinct points *A*, *B* and *C* with position vectors $\vec{a}$, $\vec{b}$ and $\vec{c}$ are collinear if and only if there exist real numbers *x,y,z,* none of them is zero, such that

$$x + y + z = 0 \text{ and } x\vec{a} + y\vec{b} + z\vec{c} = \vec{0}.$$

Example 8.3

Let *A* and *B* be two points with position vectors $2\vec{a} + 4\vec{b}$ and $2\vec{a} - 8\vec{b}$. Find the position vectors of the points which divide the line segment joining *A* and *B* in the ratio 1:3 internally and externally.

Solution

Let *O* be the origin. It is given that

$$\overrightarrow{OA} = 2\vec{a} + 4\vec{b} \text{ and } \overrightarrow{OB} = 2\vec{a} - 8\vec{b}.$$

Let *C* and *D* be the points which divide the segment *AB* in the ratio 1 : 3 internally and externally respectively. Then

$$\overrightarrow{OC} = \frac{3\overrightarrow{OA} + \overrightarrow{OB}}{3+1} = \frac{3(2\vec{a} + 4\vec{b}) + (2\vec{a} - 8\vec{b})}{4} = 2\vec{a} + \vec{b}.$$

$$\overrightarrow{OD} = \frac{3\overrightarrow{OA} - \overrightarrow{OB}}{3-1} = \frac{3(2\vec{a} + 4\vec{b}) - (2\vec{a} - 8\vec{b})}{2} = 2\vec{a} + 10\vec{b}.$$

Let us recall the definition that the line joining a vertex of a triangle with the midpoint of its opposite side is called a median. The centroid divides the median from vertex to the midpoint of the opposite side internally in the ratio 2:1.

Theorem 8.3

The medians of a triangle are concurrent.

Proof

Let *ABC* be a triangle and let *D, E, F* be the mid points of its sides *BC, CA* and *AB* respectively. We have to prove that the medians *AD, BE, CF* are concurrent.

Let *O* be the origin and $\vec{a}, \vec{b}, \vec{c}$ be the position vectors of *A, B,* and *C* respectively.

The position vectors of *D, E,* and *F* are respectively

$$\frac{\vec{b}+\vec{c}}{2}, \frac{\vec{c}+\vec{a}}{2}, \frac{\vec{a}+\vec{b}}{2}.$$

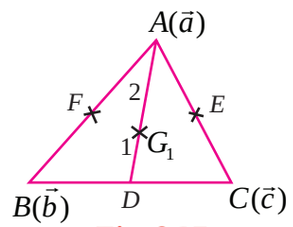


Fig. 8.27

Let G_1 be the point on *AD* dividing it internally in the ratio 2 : 1

$$\text{Therefore, position vector of } G_1 = \frac{1\overrightarrow{OA} + 2\overrightarrow{OD}}{1+2}$$

$$\overrightarrow{OG_1} = \frac{1\vec{a} + 2\left(\frac{\vec{b}+\vec{c}}{2}\right)}{3} = \frac{\vec{a}+\vec{b}+\vec{c}}{3} \quad (1)$$

Let G_2 be the point on *BE* dividing it internally in the ratio 2 : 1

$$\text{Therefore, } \overrightarrow{OG_2} = \frac{1\overrightarrow{OB} + 2\overrightarrow{OE}}{1+2}$$

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